Impact of Artemether-Lumefantrine on Oxidative Stress Markers in Aspirin-Induced Gastric Ulcerated Rat

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ABSTRACT

Artemether–Lumefantrine (AL), the first-line artemisinin-based combination therapy (ACT) for the treatment of uncomplicated malaria, exerts its antimalarial effect partly through the generation of reactive oxygen species. Although this mechanism is beneficial for parasite clearance, it may also induce oxidative stress in host tissues. This study investigated the effects of AL on oxidative stress markers and gastric ulcer index in aspirin-induced gastric ulceration in Wistar rats. Twenty adult male Wistar rats were randomly assigned to four groups (n = 5 per group): normal control, AL-treated, ulcer control, and ulcer + AL-treated groups. Gastric ulceration was induced with aspirin (300 mg/kg) following overnight fasting. Artemether–Lumefantrine (2.3/27.4 mg/kg) was administered orally once daily for three consecutive days to the appropriate groups after ulcer induction. Gastric tissues were harvested for the determination of malondialdehyde (MDA), reduced glutathione (GSH), and ulcer index, followed by histopathological examination. Data were analysed using one-way analysis of variance followed by Tukey's post hoc test, with statistical significance set at $P < 0.05$. AL-treated ulcerated rats showed significantly increased MDA levels and ulcer indices, together with significantly reduced GSH concentrations, compared with the normal control group ($P < 0.05$). Histopathological examination revealed marked inflammatory cell infiltration, epithelial disruption, and tissue damage in the gastric mucosa of ulcerated rats treated with AL. These findings suggest that AL may enhance oxidative stress and aggravate gastric mucosal injury in the presence of pre-existing ulceration under the experimental conditions of this study.

Keywords: Artermither-Lumefantrine, Oxidative Stress, Malondialdehyde, Reduced Gluthathione.

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1. INTRODUCTION

Peptic ulcer disease (PUD) is a common gastrointestinal disorder characterized by mucosal lesions affecting the stomach or proximal duodenum. Despite advances in diagnosis and treatment, it remains a significant cause of morbidity and mortality worldwide, particularly in low- and middle-income countries [1]. Gastric and duodenal ulcers constitute the major forms of PUD and result from an imbalance between aggressive factors, including gastric acid, pepsin, non-steroidal anti-inflammatory drugs (NSAIDs), *Helicobacter pylori* infection, and reactive oxygen species (ROS), and the protective mechanisms of the gastric mucosa [2].

The gastric mucosa serves as an important defensive barrier composed of a mucus-bicarbonate layer overlying the glandular epithelium. Maintenance of mucosal integrity depends on adequate blood flow, continuous epithelial regeneration, prostaglandin synthesis, mucus and bicarbonate secretion, and an effective antioxidant defence system [3, 4]. Disruption of these protective mechanisms increases susceptibility to gastric injury and ulcer formation.

Oxidative stress plays a central role in the pathogenesis of gastric ulceration. It results from an imbalance between the generation of ROS and the capacity of endogenous antioxidant systems to neutralize them.

Excessive ROS production initiates lipid peroxidation, damages cellular membranes, and promotes inflammatory responses that aggravate mucosal injury [5–9]. Malondialdehyde (MDA), a major end-product of lipid peroxidation, is widely used as a biomarker of oxidative damage, whereas reduced glutathione (GSH) represents one of the principal intracellular antioxidants responsible for maintaining cellular redox balance by scavenging free radicals and detoxifying reactive oxygen intermediates [10, 11]. Depletion of GSH and increased MDA concentrations therefore reflect oxidative injury and impaired antioxidant defence during gastric ulceration [12].

Malaria remains a major public health challenge in tropical and subtropical regions and is caused by *Plasmodium* parasites transmitted through the bites of infected female *Anopheles* mosquitoes [13]. Artemisinin-based combination therapies (ACTs) are currently recommended as the first-line treatment for uncomplicated *Plasmodium falciparum* malaria because of their high efficacy and their ability to delay the development of parasite resistance [14–18]. Among these, AL is the most widely prescribed ACT. Artemether exerts its antimalarial activity through activation by free haem released during haemoglobin digestion within the parasite, resulting in the production of reactive oxygen species and carbon-centred free radicals that damage essential parasite proteins and membranes [19, 20]. Lumefantrine complements this action by inhibiting haem detoxification, thereby enhancing parasite killing [21].

Although ROS generation is fundamental to the antimalarial efficacy of AL, increasing evidence suggests that excessive oxidative stress may also affect host tissues [12, 22–24]. This raises concerns regarding the safety of AL in patients with pre-existing gastric mucosal injury, particularly where oxidative stress already contributes to disease progression. However, information regarding the effects of AL on gastric oxidative stress and antioxidant defence during established gastric ulceration remains limited. Therefore, this study investigated the effects of Artemether–Lumefantrine on gastric oxidative stress by evaluating ulcer index, MDA, GSH, and histopathological changes in aspirin-induced gastric ulcerated Wistar rats.

2. MATERIALS AND METHODS

2.1 Experimental animals and study design: Twenty (20) adult male Wistar rats weighing 100–120 g were used for this study. The animals were obtained from the Animal House of the College of Health Sciences, Obafemi Awolowo University, Ile-Ife, Nigeria, and housed in the Animal House of the Department of Human Physiology, Federal University of Technology, Akure, Nigeria. Only male rats were used to eliminate the possible influence of hormonal fluctuations associated with the oestrous cycle on oxidative stress and metabolic responses.

Following one week of acclimatisation, the animals were randomly allocated into four experimental groups (n = 5 per group) using a simple randomisation method as previously described [25]. During the study, the rats were maintained under standard laboratory conditions at a temperature of 22–25°C, relative humidity of 50–60%, and a 12-hour light/dark cycle. They were housed in clean, well-ventilated cages with adequate spacing and were provided with standard commercial rat pellets (Ladoke Feeds, Ibadan, Nigeria) and clean drinking water *ad libitum*.

Animal handling and experimental procedures were conducted in accordance with the National Institutes of Health (NIH) Guide for the Care and Use of Laboratory Animals. Every effort was made to minimise animal discomfort throughout the study. The experimental groups were as follows:
Group I (Control): Normal control rats receiving distilled water only.

Group II (AL): Rats administered Artemether–Lumefantrine only.
Group III (Ulcer + AL): Rats with aspirin-induced gastric ulcer treated with Artemether–Lumefantrine.
Group IV (Ulcer Control): Rats with aspirin-induced gastric ulcer receiving normal saline.

2.2 Induction of gastric ulcer: Gastric ulceration was induced according to the method previously described [26]. Following an overnight fast with free access to water, gastric ulcers were induced in Groups III and IV by oral administration of aspirin (300 mg/kg body weight) suspended in 1% carboxymethylcellulose (CMC) using an oral gavage for three consecutive days. Food was withheld throughout the ulcer induction period, while water was provided *ad libitum*.

2.3 Artemether–Lumefantrine treatment: Following ulcer induction, Artemether–Lumefantrine (Combiart®, 2.3/27.4 mg/kg body weight) was administered orally once daily for three consecutive days to Groups II and III. The selected dose was based on a previously published experimental study [27]. Group IV (ulcer control) received normal saline (1 mL/kg body weight) orally over the same treatment period. All drugs and reagents were obtained from a licensed pharmaceutical outlet in Akure, Ondo State, Nigeria.

Table 1. Experimental Grouping and Treatment Protocol

Group	Treatment
Group I	Normal control; no ulcer induction and no Artemether–Lumefantrine treatment
Group II	Artemether–Lumefantrine only; no ulcer induction
Group III	Aspirin-induced gastric ulcer followed by Artemether–Lumefantrine treatment
Group IV	Aspirin-induced gastric ulcer only (ulcer control)

2.4 Determination of gastric ulcer index: Twenty-four hours after the final treatment, the animals were euthanised by cervical dislocation. The abdomen was opened through a midline incision, and the stomach was carefully excised and opened along the greater curvature. Gastric contents were gently removed with normal saline, and the stomach was pinned flat on a wax board for examination.

Ulcer lesions were identified using a ×10 magnifying lens. Black or red haemorrhagic spots and linear lesions along the gastric mucosa were recorded as ulcers. The length of each lesion was measured using a divider and ruler. The ulcer index for each animal was calculated as the sum of all ulcer lengths, with each haemorrhagic spot assigned a value of 0.5 mm. Measurements were performed twice, and the mean value was used for analysis [26].

2.5 Determination of reduced glutathione (GSH): GSH concentration in gastric mucosal tissue was determined using the method previously described [28]. Approximately 1 g of gastric mucosal tissue was homogenised, and the homogenate was centrifuged at 3,000 rpm for 10 minutes at 4°C. The resulting supernatant was reacted with 5,5′-dithiobis-(2-nitrobenzoic acid) (DTNB), and absorbance was measured colorimetrically at 520 nm. GSH concentration was calculated from a standard calibration curve and expressed as μmol/g of gastric tissue.

2.6 Determination of Malondialdehyde (MDA): Lipid peroxidation was assessed by measuring gastric mucosal MDA according to the method previously described [29]. Approximately 1 g of gastric mucosal tissue was homogenised in Tris-HCl buffer containing butylated hydroxytoluene to prevent further oxidation. The homogenate was centrifuged at 3,000 rpm for 10 minutes at 4°C to obtain the supernatant. The supernatant was reacted with N-methyl-2-phenylindole, incubated at 45°C, and absorbance was measured at 540 nm using a Sherwood® Model

254 colorimeter. MDA concentration was calculated from a standard curve and expressed as μmol/g of gastric tissue.

2.7 Statistical analysis: Data were analysed using GraphPad Prism (2019 version). Results are presented as mean ± standard deviation (SD). Data were first assessed for normality and homogeneity of variance before statistical comparisons. Differences among groups were analysed using one-way analysis of variance (ANOVA), followed by Tukey's multiple comparison post hoc test where appropriate. Statistical significance was accepted at $P < 0.05$.

3. RESULTS

3.1 Effects of Artemether–Lumefantrine on gastric ulcer index, MDA and GSH levels

Figure 1 illustrates the effects of AL on gastric ulcer index, MDA, and GSH levels in the experimental groups. The ulcer control and ulcer + AL-treated groups exhibited significantly higher ulcer indices than the normal control and AL-treated groups ($P < 0.05$). Although the ulcer + AL-treated group showed a higher ulcer index than the ulcer control group, this difference was not statistically significant ($P > 0.05$).

MDA concentrations were higher in the AL-treated and ulcer control groups than in the normal control group; however, these increases were not statistically significant ($P > 0.05$). In contrast, the ulcer + AL-treated group demonstrated a significant increase in MDA levels compared with the normal control group ($P < 0.05$).

GSH concentrations were lower in the ulcer control and ulcer + AL-treated groups than in the normal control group. The ulcer + AL-treated group showed a significant reduction in GSH levels compared with the normal control group ($P < 0.05$). No significant difference in GSH concentration was observed between the ulcer + AL-treated group and either the AL-treated or ulcer control groups ($P > 0.05$).

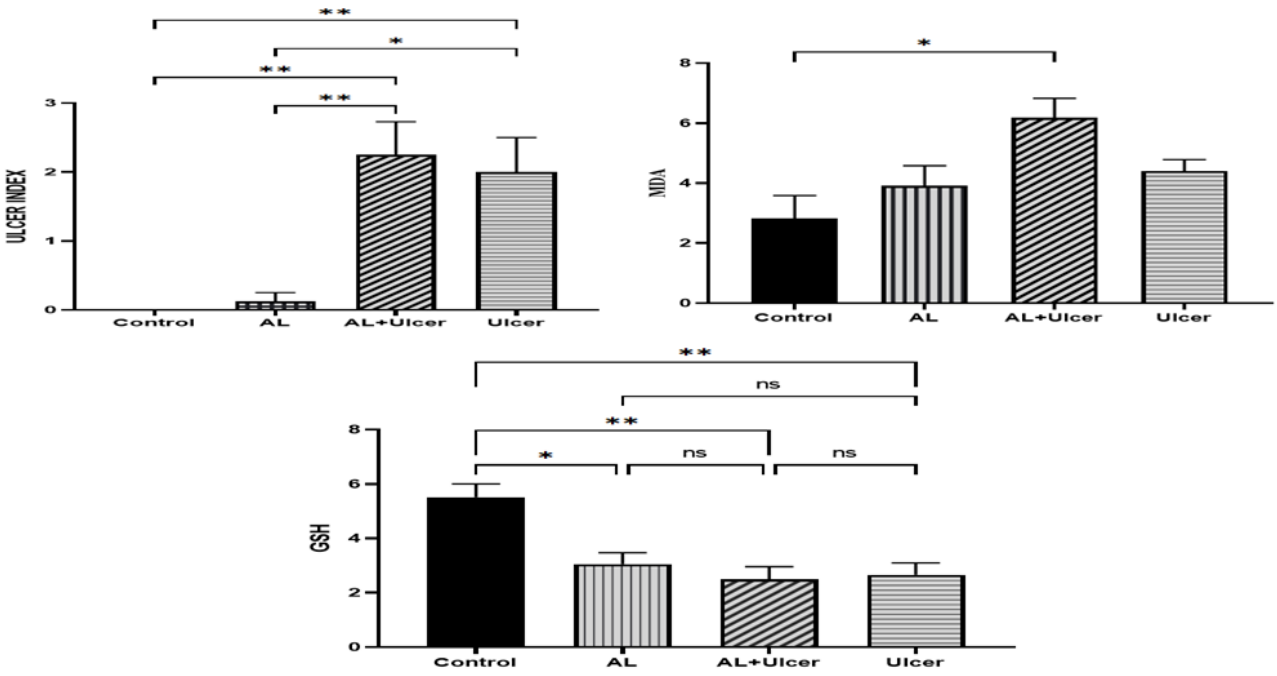


Figure 1. Effects of Artemether–Lumefantrine on (A) gastric ulcer index, (B) malondialdehyde (MDA), and (C) reduced glutathione (GSH) levels in experimental rats.
Data are presented as mean ± SD (n = 5 per group). *Statistical significance was accepted at $P < 0.05$.

3.2 Histopathological findings of gastric mucosa

Representative photomicrographs of gastric mucosal sections are presented in Figure 2. The gastric mucosa of the normal control and AL-treated rats exhibited normal histological architecture, with intact mucosal and submucosal layers and no evidence of ulceration or inflammatory cell infiltration.

The ulcer control group showed severe gastric mucosal injury characterised by disruption of the epithelial lining, deep ulcer craters extending into the submucosa, marked vascular congestion with haemorrhage, and dense inflammatory cell infiltration, predominantly by neutrophils.

Similarly, the ulcer + AL-treated group exhibited pronounced histopathological alterations, including epithelial ulceration, distortion of the mucosal architecture, and mild-to-moderate inflammatory cell infiltration. Histological evidence of gastric mucosal injury was more pronounced than in the normal control and AL-treated groups.

4. DISCUSSION

The present study investigated the effects of AL on gastric oxidative stress and mucosal injury in aspirin-induced gastric ulcerated Wistar rats. The principal findings demonstrate that AL administration in ulcerated rats was associated with an increase in gastric ulcer index and MDA levels, together with a reduction in GSH concentrations, accompanied by histopathological evidence of gastric mucosal injury. These findings suggest that AL may aggravate oxidative stress and gastric mucosal damage in the presence of pre-existing ulceration.

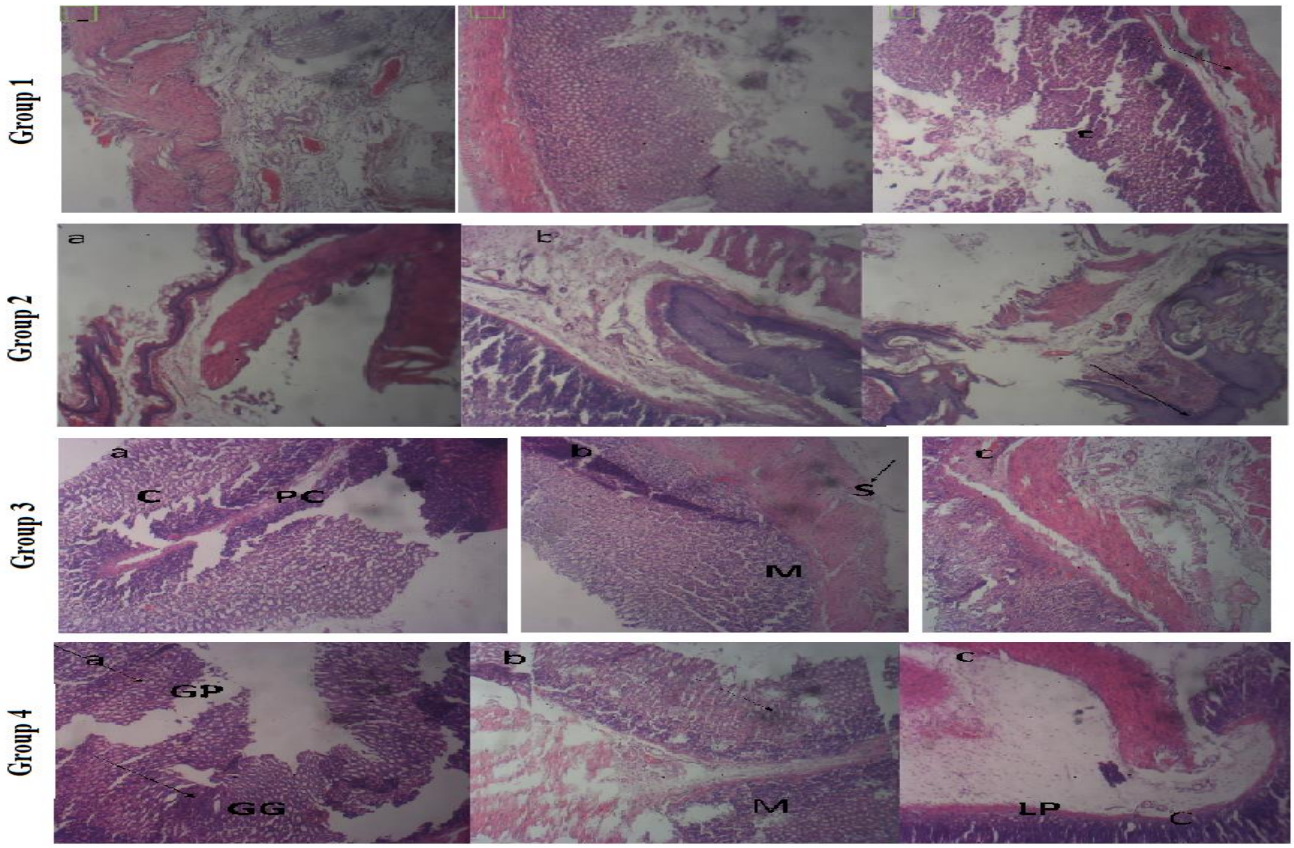


Figure 2. Photomicrograph of gastric mucosa histopathological lesions (H&E x 400).

The ulcer index observed in the ulcer + AL-treated group was higher than that of the ulcer control group, although AL administration alone did not produce significant gastric mucosal injury. This finding suggests that AL is unlikely to induce gastric ulceration in normal gastric mucosa but may exacerbate existing gastric lesions under conditions of compromised mucosal integrity. Similar observations have been reported for several antimalarial agents, including amodiaquine, quinine, and chloroquine, which have been shown to influence gastric mucosal integrity through oxidative mechanisms [27, 30–32]. The present findings therefore support the hypothesis that oxidative stress contributes to the worsening of gastric injury in ulcerated animals receiving AL.

Oxidative stress is recognised as an important mechanism in the pathogenesis of gastric ulceration. Excessive production of ROS promotes lipid peroxidation, disrupts cellular membranes, impairs epithelial regeneration, and amplifies inflammatory responses, thereby delaying ulcer healing [8, 9]. In the present study, ulcerated rats treated with AL exhibited significantly higher MDA concentrations than the normal control group. Since MDA is a well-established marker of lipid peroxidation, these findings indicate increased oxidative damage to gastric tissues. Although MDA levels in the ulcer + AL-treated group were not significantly higher than those of the ulcer-only group, the observed upward trend suggests that AL may contribute to oxidative injury in an already compromised gastric environment.

The increase in oxidative stress observed in this study is biologically plausible considering the pharmacological mechanism of Artemether–Lumefantrine. Artemether undergoes activation following interaction with free haem released during haemoglobin digestion by *Plasmodium* parasites, generating reactive oxygen species and carbon-centred free radicals that damage parasite proteins and membranes [20, 35]. Lumefantrine complements this action by inhibiting haem detoxification, thereby increasing oxidative stress within the parasite [21]. Although these mechanisms are essential for parasite clearance, excessive ROS generation may also affect host tissues, particularly under pathological conditions associated with impaired antioxidant defence.

GSH is one of the principal intracellular antioxidants responsible for maintaining cellular redox homeostasis by scavenging reactive oxygen species and protecting cellular components from oxidative damage. In the present study, GSH concentrations were significantly reduced in ulcerated rats treated with AL compared with the normal control group, indicating depletion of endogenous antioxidant reserves. Similar reductions in GSH have been reported following exposure to oxidative stress-inducing antimalarial therapies [22, 24]. The concomitant increase in MDA and reduction in GSH observed in this study provide complementary evidence of oxidative imbalance in gastric tissue following AL administration in ulcerated rats.

Histopathological examination further supported the biochemical findings. While the gastric mucosa of the normal control and AL-treated groups remained largely intact, ulcerated animals demonstrated epithelial disruption, inflammatory cell infiltration, vascular congestion, haemorrhage, and distortion of mucosal architecture. These structural changes are consistent with oxidative tissue injury and inflammatory responses commonly observed in experimental models of gastric ulceration. The more pronounced mucosal alterations observed in ulcerated rats treated with AL suggest that oxidative stress may contribute to delayed mucosal recovery under these experimental conditions.

Although the present findings indicate that AL may aggravate oxidative injury in ulcerated gastric tissue, caution should be exercised when extrapolating these findings to clinical practice. The study was conducted in a relatively small number of male rats using an

experimentally induced ulcer model and evaluated only two oxidative stress biomarkers (MDA and GSH). Additional studies incorporating larger sample sizes, female animals, longer treatment durations, and broader assessment of antioxidant enzymes, inflammatory cytokines, prostaglandins, gastric mucus secretion, and molecular signalling pathways are required to provide a more comprehensive understanding of the effects of AL on gastric mucosal integrity.

5. CONCLUSION

The present study demonstrated that Artemether–Lumefantrine administration in aspirin-induced gastric ulcerated rats was associated with increased lipid peroxidation, depletion of reduced glutathione, and worsening of gastric mucosal injury. These findings suggest that AL may enhance oxidative stress in the presence of pre-existing gastric ulceration under the experimental conditions employed. However, the results should be interpreted cautiously, and further experimental and clinical studies are required to establish the clinical significance of these observations.

Study limitations: This study evaluated oxidative stress using only malondialdehyde and reduced glutathione as biochemical markers. Other components of the antioxidant defence system, including superoxide dismutase, catalase, glutathione peroxidase, inflammatory cytokines, prostaglandins, nitric oxide, gastric mucus secretion, mucosal blood flow, and molecular markers of oxidative injury, were not investigated. In addition, the study involved only male rats and a relatively small sample size. These limitations should be considered when interpreting the findings and should be addressed in future studies.

Conflict of interest: The authors declare that they have no conflict of interest.

Data availability statement: The datasets generated and analysed during the current study are available from the corresponding author upon reasonable request.

Authors' contributions: M.A.A. conceived and designed the study, developed the methodology, supervised the research, performed data acquisition, validated the findings, and prepared the original manuscript draft. O.O.O. performed the statistical analyses and contributed to manuscript writing, critical revision, and editing. Both authors read and approved the final version of the manuscript.

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